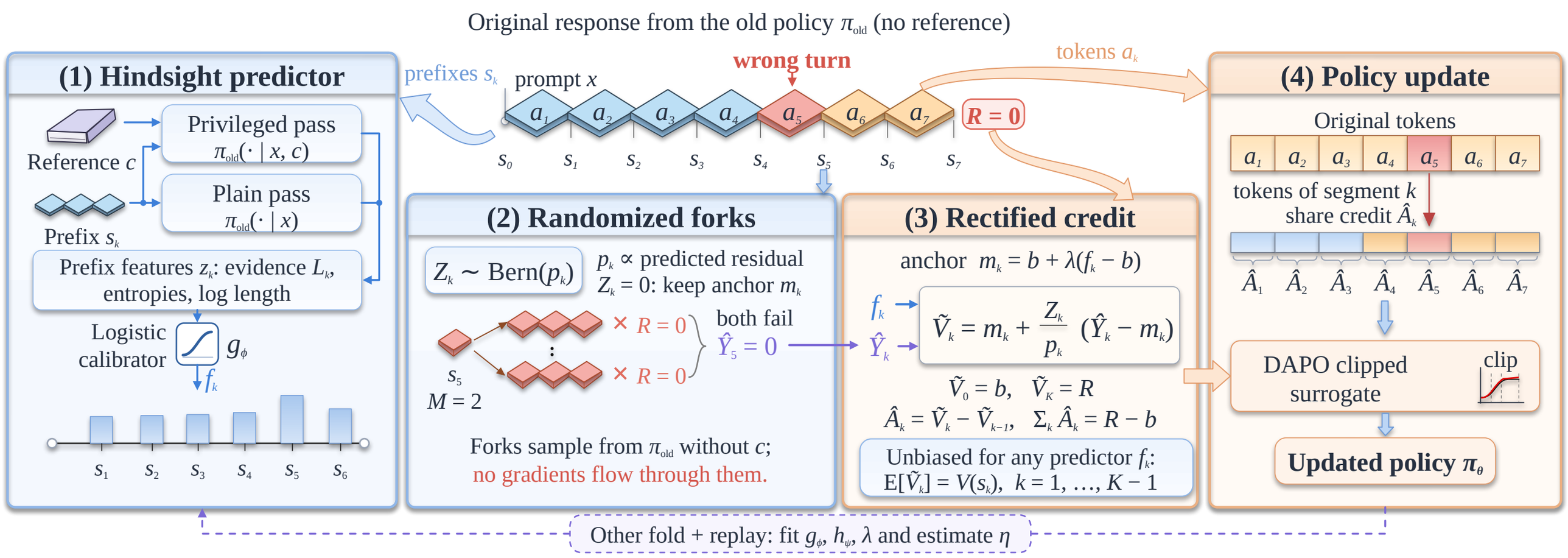

Original response from the old policy π_{old} (no reference)
(1) Hindsight predictor
Reference c
Prefix s_k
Privileged pass
$\pi_{\text{old}}(\cdot \mid x, c)$
Plain pass
$\pi_{\text{old}}(\cdot \mid x)$
Prefix features z_k: evidence L_k, entropies, log length
Logistic calibrator
g_ϕ
f_k
s_1 s_2 s_3 s_4 s_5 s_6
prefixes s_k
prompt x
wrong turn
tokens a_k
a_1 a_2 a_3 a_4 a_5 a_6 a_7
$R = 0$
s_0 s_1 s_2 s_3 s_4 s_5 s_6 s_7
(2) Randomized forks
$Z_k \sim \text{Bern}(p_k)$
$p_k \propto$ predicted residual
$Z_k = 0$: keep anchor m_k
s_5
$M = 2$
$\times$ $R = 0$
$\times$ $R = 0$
both fail
$\hat{Y}_5 = 0$
Forks sample from π_{old} without c;
no gradients flow through them.
(3) Rectified credit
anchor $m_k = b + \lambda(f_k - b)$
f_k
$\hat{Y}_k$
$\tilde{V}_k = m_k + \frac{Z_k}{p_k}(\hat{Y}_k - m_k)$
$\tilde{V}_0 = b, \quad \tilde{V}_K = R$
$\hat{A}_k = \tilde{V}_k - \tilde{V}_{k-1}, \quad \Sigma_k \hat{A}_k = R - b$
Unbiased for any predictor f_k:
$\mathrm{E}[\tilde{V}_k] = V(s_k), \ k = 1, \ldots, K-1$
(4) Policy update
Original tokens
a_1 a_2 a_3 a_4 a_5 a_6 a_7
tokens of segment k
share credit $\hat{A}_k$
$\hat{A}_1$ $\hat{A}_2$ $\hat{A}_3$ $\hat{A}_4$ $\hat{A}_5$ $\hat{A}_6$ $\hat{A}_7$
DAPO clipped surrogate
clip
Updated policy π_θ
Other fold + replay: fit g_ϕ, h_ψ, λ and estimate η